

JECRC UNIVERSITY, JAIPUR

Session: Jan-June 2026

Subject: Mathematics-II (DMA025A)

B. Tech. II Semester I Year 2025-26

Tutorial Sheet-15 (UNIT-5)

M.M =64

PART – A

(5*2=10)

A1: Define Maximum Modulus Theorem.

A2: Define non- isolated singularity of a complex function.

A3: Define Isolated Singularity and Simple Pole.

A4: State Liouville's Theorem.

A5: Find the zeroes of the function $f(z) = (z + i)^2(z - i)^2$ and state the order.

PART – B

(3*7=21)

B1: Evaluate $\oint \frac{3z^2+z}{z^2-1} dz$, where C is the circle $|z - 1| = 1$.

B2: Expand $f(z) = \cos z$ in Taylor's series about $z = \frac{\pi}{4}$.

B3: Let $f(z) = 10 - 3i$. Determine whether $f(z)$ satisfies the condition of Liouville's theorem and state the nature of the function.

PART – C

(3*11=33)

C1: Determine the poles of the following function and residue at each pole $\oint_0^{2\pi} \frac{1-2z}{z(z-1)(z-2)}$

$|z| = 1.5$.

C2: Evaluate $\frac{1}{2\pi i} \oint \frac{z^2}{z^2+4} dz$, where C is the square with vertices at +2,-2,+4i,-4i.

C3: Evaluate the integral $f(z) = \int_0^{2\pi} \frac{d\theta}{2+\cos \theta}$.

Answer:

Part A:

A5: $Z=i$ of order 2 and $z=-i$ of order 2.

Part B:

B1: $8\pi i$ B2: $\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\left(z - \frac{\pi}{4}\right) - \frac{1}{2\sqrt{2}}\left(z - \frac{\pi}{4}\right)^2 + \dots \dots \dots$

B3: $f(z)$ satisfies the condition of Liouville's Theorem because whether the function is entire and bounded function and the nature of the function is constant.

Part C:

C1: Poles=0,1,2 and *Residue* = $3\pi i$

C2: 0

C3: $\frac{2\pi}{\sqrt{3}}$